

DEVOPS INTERVIEW QUESTION

A. GIT

1. What is Git?
2. Difference between Git and GitHub?
3. What is the difference between git clone and git fork?
4. What are the stages in Git?
5. What is git merge
6. Difference between merge and rebase?
7. What is a merge conflict? and how to resolve it ?
8. What is .gitignore?
9. How is Git used in DevOps?
10. Difference between git pull and git fetch?
11. What is git reset?
12. What is HEAD in Git?
13. What is git cherry-pick?
14. Difference between git reset and git revert?
15. How do you recover a deleted commit?

Scenario Based :

1. A wrong commit pushed to main. What will you do?
2. Two developers modified the same file and a merge conflict occurred. How do you resolve it?
3. One commit from another branch is required urgently in production. What will you do?
4. Your branch has too many unnecessary commits. How do you clean history before PR?

B. LINUX AND SHELL SCRIPTING

1. Server disk is full suddenly. How will you troubleshoot?
2. High CPU usage on server. What will you check?
3. A service is down. How do you bring it up?
4. How do you find which port is used by a process?
5. File deleted accidentally. Can it be recovered?
6. User cannot log in. How do you debug?
7. How do you check server uptime?
8. Difference between soft link and hard link?
9. Difference between = and ==?
10. What is shebang?
11. Script to check file exists

12. How do you make script executable?
13. grep | awk | sed
14. ps | top | free
15. df | du
16. chmod | chown
17. tar | gzip
18. curl | wget
19. find
20. Production server crashed at midnight. What is your approach?
21. A service fails after reboot. How do you debug?
22. Linux basic commads

C. MAVEN

1. Build is failing due to corrupted dependency. How do you fix it?
2. What is meant by build automation
3. build life cycle
4. How Maven is used in CI/CD pipelines?
5. How do you skip tests in production build?
6. What is the role of pom.xml?
7. Difference between clean, install, package, deploy?
8. What happens when you run mvn install?
9. How do you integrate SonarQube with Maven?
10. Maven phases of building code

D. JENKINS

1. Jenkins job is failing but works locally. How will you troubleshoot?
2. Jenkins service is down. How do you bring it up?
3. Jenkins UI is slow. What could be the reasons?
4. Build is stuck in queue. Why?
5. Pipeline suddenly stopped triggering after Git push. Why?
6. How do you pass credentials securely in Jenkins pipeline?
7. Jenkins pipeline needs different behavior for dev and prod. How?
8. Agent is offline. How do you fix it?
9. Difference between master and agent?
10. Jenkins running in Docker. Build fails with permission denied. Why?
11. How do you take Jenkins backup?
12. what is meant by continuous integration, continuous delivery and deployment ?
13. Difference between continuous delivery and continuous deployment ?
14. what are build triggers ?
15. diffrence between freestyle and pipeline ?
16. Pipeline is working yesterday but failing today without code change. What will you check?

17. Build stuck in “Waiting for executor”. Why and how to fix?
18. Jenkins pipeline fails with “permission denied”. What do you do?
19. How do you pass different values for dev, qa, prod in Jenkins?
20. Types of pipeline
21. what is jenkinsfile ?
22. what is meant by post build actions ?
23. Master-slave architecture ?
24. Sonarqube and nexus integration with jenkins why ?

E. DOCKER

1. Container is running but application is not accessible. What will you check?
2. Container exits immediately after starting. How do you debug?
3. Docker image size is very large. How will you reduce it?
4. Container data lost after restart. Why?
5. container VS virtual machine
6. Permission denied when container writes to mounted volume. Fix?
7. Container can access internet but host cannot access container. Why?
8. Docker disk space full. How do you clean safely?
9. Difference between CMD and ENTRYPOINT ?
10. How Docker fits into CI/CD pipeline?
11. Two services need to communicate. How do you do it in Docker?
12. Docker container restarts continuously. Why?
13. How do you debug inside a running container?
14. what is docker-compose
15. write a dockerfile
16. Difference between dockerfile and multi-stage dockerfile
17. Difference between ADD and COPY ?

F. KUBERNETES

1. Pod is in CrashLoopBackOff. How do you troubleshoot?
2. Pod is running but application is not accessible. What will you check?
3. Pod stuck in Pending state. Why?
4. ImagePullBackOff error. What could be wrong?
5. Node status shows NotReady. How do you debug?
6. One node is overloaded. How do you fix it?
7. New deployment failed. How do you rollback?
8. Explain kubernetes architecture ?
9. what is control plane and its components ?
10. what is pod ?
11. Difference between Replication Controller and ReplicaSet ?
12. How do you check deployment rollout status?

13. Application downtime happened during deployment. How to avoid next time?
14. Deployment Strategies ?
15. Service is created but not reachable from browser. Why?
16. What is the difference between Service and Ingress ?
17. ConfigMap updated but app still uses old values. Why?
18. Pod fails due to volume mount error. How do you debug?
19. Data lost after pod restart. Why?
20. Traffic increased suddenly. How do you handle scaling?
21. Pods are restarting due to OOMKilled. Fix?
22. Cluster is running out of IPs. What is the reason?
23. Pods cannot talk to each other. Why?
24. PVC stuck in Pending. What are the reasons?
25. Developer says: "I get Forbidden error when running kubectl get pods". What is the issue?
26. You want a user to access pods ONLY in dev namespace. How will you do it?
27. Difference between Role and ClusterRole
28. What is daemon set ?
29. Difference between Deployment and Statefulset ?
30. What is meant by Namespaces ?

G. TERRAFORM

1. Terraform apply worked yesterday but fails today without code change. What will you check?
2. What is meant by IAAC ?
3. Terraform shows changes even though nothing was modified. Why?
4. Terraform apply failed halfway. Infrastructure is partially created. What will you do?
5. You want to separate dev, qa, prod infrastructure. How do you design Terraform?
6. Same Terraform code repeated for multiple environments. How do you optimize?
7. How do you pass different values for different environments?
8. Someone manually deleted cloud resources. Terraform still thinks they exist. Fix?
9. Terraform destroy executed accidentally. How do you prevent this next time?

H. ANSIBLE

1. Ansible playbook works on one server but fails on others. Why?
2. Ansible shows "UNREACHABLE". How do you debug?
3. Playbook ran successfully but changes didn't apply. Why?
4. Task executed on wrong servers. What went wrong?
5. What is meant by Playbook ?
6. How do you manage different environments (dev/qa/prod)?
7. Playbook failed in the middle. How do you resume?
8. You want to run only one task from a playbook. How?
9. Same playbook repeated across projects. How do you optimize?

10. Variable value is not picking correctly. Why?
11. Passwords stored in plain text in playbook. What is the fix?
12. Playbook is slow. How do you optimize?